

SUFIYAN KHAN

Gen AI/ AI - ML / HonoursINDataScience/GoaCollegeOf Engineering

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WORK EXPERIENCES

SoftwareDevelopment | Internship

Jul2025 - Sep2025

Civil Supplies Department

Hybrid•Panaji, Goa

- Built road-aware **FPS Redirection System** serving **1,400+ shops** and **8 lakh+ households** across Goa using OpenRouteService API for real road distances
- Developed **Smart Redirection algorithm** reassigning **41+ beneficiaries per operation**, replacing straight-line estimates with actual road routing
- Implemented **real-time FPS Status Management** for **43 shops** with instant Open/Closed toggling triggering automatic beneficiary reassignment
- Integrated **Leaflet.js Map Visualization** showing user location, closed FPS, top-3 alternatives, and route paths across **North & South Goa**
- Built **Admin Dashboard** for managing FPS network across **2 districts** with downloadable Excel reports for audit and oversight
- Deployed on **Vercel** with secure auth, processing bulk Excel uploads of **43 FPS** and **41+ beneficiary records** per reassignment cycle

Technologies / Skills Used:Node.js, React, Leaflet.js, OpenRouteService API, Bootstrap, JavaScript, GitHub, Vercel

PROJECTS

Self-Improving RAG (Retrieval-Augmented Generation) | 🌐 Website

Present

AI/ML Engineer

- Built **FAISS + BM25 Hybrid Retrieval** via Reciprocal Rank Fusion + **Cross-Encoder reranking**, improving retrieval precision by **~30%** over single-method approaches
- Integrated **4 LLM providers** (Gemini, LLaMA 3/Groq, NVIDIA, Ollama) with **LangSmith tracing** capturing **100%** of pipeline runs with latency metadata
- Designed **Smart Feedback Loop** collecting **50+ signals** (thumbs up/down, dwell time, citation clicks) to auto-build Cross-Encoder training datasets
- Built **automated fine-tuning pipeline** with background scheduler, evaluating via **NDCG@K** — deploys only if new reranker outperforms baseline
- Developed **A/B testing framework** comparing reranker versions, ensuring only performance-validated models are promoted to production
- Built **Streamlit UI** with chat, document ingestion, and real-time metrics backed by async **SQLite** — **0 manual intervention** in retraining cycle

Technologies / Tools Used:Python, FAISS, BM25, SentenceTransformers, Groq, LangSmith, SQLite, Streamlit

Aura Voice Agent | 🌐 Website

Present

Agentic AI Developer

- Built **real-time voice AI assistant** using Python async WebSocket + React, supporting natural conversation with **2 LLMs** (Gemma 4 31B & Qwen 3.5 122B) via Ollama Cloud
- Implemented **persistent memory** using PostgreSQL across **4 data stores** — conversation history, user profile, alarms, and documents — surviving restarts and redeployments
- Developed **RAG pipeline** using HuggingFace **BAAI/bge-small-en-v1.5** embeddings + ChromaDB vector search, enabling voice-based chat with uploaded PDFs with **0 manual indexing**
- Integrated **Smart Tool Routing** across **6+ tools** — Weather, Calculator, Alarms, Web Search (Tavily), RAG, Gmail API — auto-invoked based on user intent
- Added **3-language support** (English, Hindi, Hinglish) using Web Speech API (STT) and Browser Speech Synthesis (TTS) with smart language matching
- Deployed on **Cloudflare Pages + Railway** via secure WebSocket (wss://), with **100% observability** via Langfuse tracing across all pipeline runs

Technologies / Tools Used:Python, React, PostgreSQL, Ollama, ChromaDB, Tavily, WebSockets, Langfuse

SKILLS

Databases :	PostgreSQL, MySQL, SQLite
Programming Languages:	Python, Java, SQL
AI ML:	PyTorch, Scikit-learn, Transformers, Deep Learning, CNNs, RNNs, LSTMs, Embeddings, Fine-tuning, RAG, Prompt Engineering, Machine Learning
Llm Frameworks:	LangChain, LangGraph, LlamaIndex, SentenceTransformers
Vector Databases & Retrieval:	FAISS, ChromaDB, BM25, Hybrid Search, Reranking
Deployment & Tools:	Docker, Git, GitHub, Streamlit, Render, Railway, Ollama, LangSmith, Langfuse
Libraries :	Pandas, NumPy, Matplotlib, OpenCV

EDUCATIONS

GoaEngineeringCollege,Ponda|Ponda, Goa

Sep 2022 - Present

Bachelor in Information Technology

CGPA : 7.55

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CERTIFICATIONS

IBM- Machine Learning with Python
IBM- Databases and SQL for DataScience
IBM- Python for Data Science, AI and Development
Nptel- Data Analytics with Python
Nptel - Joy of computing using python
HuggingFace - Agent Course